

ABHINAV SINGH

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EXPERIENCE

AlertEnterprise

Sep. 2021 – Apr. 2023

Software Engineer I

Remote

- Developed high-performance RESTful APIs in Spring Boot to streamline operations across 50+ airports.
- Engineered a Kubernetes-based microservices platform with distributed authentication, leveraging Istio for secure inter-service communication and fine-grained RBAC to drive scalable deployments and reduce vulnerabilities.
- Optimized HQL queries and refactored Java data access layers by enhancing indexing and eliminating redundant joins, boosting performance by 21% in 3 months.
- Built a policy enforcement engine using rule-based decision logic with asynchronous event processing and caching, cutting workflow time by 15%.
- Enhanced Angular product screens by implementing efficient code splitting and lazy loading, reducing API calls and slashing page load times by 20% to deliver a faster, more responsive UI.
- Collaborated in an agile environment using JIRA to streamline documentation and QA.

California State University, Fullerton

Jan. 2025 – Present

Graduate Teaching Associate

Fullerton, CA

- Teach Data Structures and Algorithms to a class of 38 students, designing algorithmic problem-solving sessions, coding challenges, and optimizing solutions using time and space complexity analysis.

PROJECTS

Enrollment API | Python, FastAPI, Krakend, DynamoDB

Aug. 2024 – Dec. 2024

- Engineered a scalable, serverless backend for college enrollment using FastAPI on AWS Lambda with API Gateway; integrated asynchronous processing and CI/CD testing to reduce API latency by 30%.
- Designed a secure role-based authentication system with JWT via Krakend; applied dependency injection and repository patterns for DynamoDB Streams and LiteFS abstraction.
- Optimized waitlist management using Redis Sorted Sets and RabbitMQ; integrated AWS SES and webhooks to reliably notify 1000+ students under heavy concurrency.

CareerRefined: Resume Maker | FastAPI, React, AWS, NLP

May. 2024 – July. 2024

- Leveraged HuggingFace's BERT with TensorFlow to build an AI resume engine that accurately extracts candidate skills, validated with PyTest to enhance job matching.
- Performed integration of a PostgreSQL-backed job tracking dashboard on AWS RDS; containerized with Docker and deployed via Elastic Beanstalk to ensure robust data synchronization.
- Deployed a responsive resume scoring UI with React, Redux, Tailwind, and Axios that processes real-time data to compute live scores; validated with Jest and Enzyme, boosting matching efficiency by 22%.

Reliable File Transfer System | C, POSIX Sockets, Threads

Nov. 2023 – Dec. 2023

- Crafted a UDP file transfer system for local networks by designing a custom sliding-window protocol with packet sequencing, CRC detection, and ACK handling via POSIX sockets, ensuring reliable delivery.
- Implemented adaptive retransmission and congestion control using POSIX threads, monitoring latency and packet loss to optimize buffer management and error recovery for high-throughput transfers.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, TypeScript, HTML/CSS, JSON, XML

Frameworks: SpringBoot, Spring MVC, Flutter, FastAPI, React, Angular, Tailwind, Node.js, JUnit

Developer Tools: Git, Google Cloud Platform, VS Code, Postman, Fiddler, Eclipse

Cloud Technologies: AWS EC2, AWS RDS, AWS S3, AWS SES, Docker, Kubernetes

DBS Applications: JDBC, MongoDB, Redis, PostgreSQL

Other Skills & Tools: Object-Oriented Programming, Maven, Gradle, Git, Apache Tomcat

EDUCATION

California State University, Fullerton

Aug. 2023 – May. 2025

Master of Science in Computer Science

Fullerton, CA